```
lab-03-15@lab-03-15-OptiPlex-3280-AIO:~$ vim prio_pree_1288.c
lab-03-15@lab-03-15-OptiPlex-3280-AIO:~$ gcc prio_pree_1288.c
lab-03-15@lab-03-15-OptiPlex-3280-AIO:~$ ./a.out
Enter number of processes: 3
Enter Arrival Time, Burst Time and Priority:
P1 AT: 2
P1 BT: 4
P1 Priority: 1
P2 AT: 5
P2 BT: 3
P2 Priority: 2
P3 AT: 0
P3 BT: 3
P3 Priority: 3

Gantt Chart:
| P3 | P1 | P2 | P3 |

Process AT      BT      PR      CT      WT      TAT
P1      2       4       1       6       0       4
P2      5       3       2       9       1       4
P3      0       3       3       10      7       10

Average WT = 2.67
Average TAT = 6.00
lab-03-15@lab-03-15-OptiPlex-3280-AIO:~$
```

```
        for(int i = 0; i < n; i++) {
            if(at[i] <= time && completed[i] == 0) {
                if(pr[i] < highest_pr) {
                    highest_pr = pr[i];
                    idx = i;
                }
            }
        }
    }
    if(idx != -1) {
        gantt_p[g] = idx;
        time += bt[idx];
        g++;
        gantt_t[g] = time;

        ct[idx] = time;
        tat[idx] = ct[idx] - at[idx];
        wt[idx] = tat[idx] - bt[idx];

        total_wt += wt[idx];
        total_tat += tat[idx];

        completed[idx] = 1;
        done++;
    } else {
        time++;
    }
}

// Print Gantt Chart
printf("\nGantt Chart:\n|");
for(int i = 0; i < g; i++)
    printf(" P%d |", gantt_p[i] + 1);

printf("\n%d", gantt_t[0]);
for(int i = 1; i <= g; i++)
```

```c
#include <stdio.h>
#include <limits.h>

int main() {
    int n;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    int at[n], bt[n], pr[n];
    int ct[n], wt[n], tat[n], completed[n];
    int gantt_p[100], gantt_t[100];

    for(int i = 0; i < n; i++)
        completed[i] = 0;

    printf("Enter Arrival Time, Burst Time and Priority:\n");
    for(int i = 0; i < n; i++) {
        printf("P%d AT: ", i+1);
        scanf("%d", &at[i]);
        printf("P%d BT: ", i+1);
        scanf("%d", &bt[i]);
        printf("P%d Priority: ", i+1);
        scanf("%d", &pr[i]);
    }

    int time = 0, done = 0, g = 0;
    float total_wt = 0, total_tat = 0;

    gantt_t[g] = 0;

    while(done < n) {
        int idx = -1;
        int highest_pr = INT_MAX;

        for(int i = 0; i < n; i++) {
            if(at[i] <= time && completed[i] == 0) {
                if(pr[i] < highest_pr) {
```

```
lab-03-15@lab-03-15-OptiPlex-3280-AIO:~$ vim prio_Non_pree_1288.c
lab-03-15@lab-03-15-OptiPlex-3280-AIO:~$ gcc prio_Non_pree_1288.c
lab-03-15@lab-03-15-OptiPlex-3280-AIO:~$ ./a.out
Enter number of processes: 3
Enter Arrival Time, Burst Time and Priority:
P1 AT: 2
P1 BT: 4
P1 Priority: 3
P2 AT: 5
P2 BT: 5
P2 Priority: 2
P3 AT: 1
P3 BT: 5
P3 Priority: 3

Gantt Chart:
| P3 | P2 | P1 |
0    6    11    15

Process AT      BT      PR      CT      WT      TAT
P1      2       4       3       15      9       13
P2      5       5       2       11      1       6
P3      1       5       3       6       0       5

Average WT = 3.33
Average TAT = 8.00
lab-03-15@lab-03-15-OptiPlex-3280-AIO:~$
```

```c
            }
        }

        if(idx != -1) {
            if(prev != idx) {
                printf(" P%d |", idx+1);
                prev = idx;
            }

            rt[idx]--;
            time++;

            if(rt[idx] == 0) {
                completed++;
                ct[idx] = time;
                tat[idx] = ct[idx] - at[idx];
                wt[idx] = tat[idx] - bt[idx];

                total_wt += wt[idx];
                total_tat += tat[idx];
            }
        } else {
            time++;
        }
    }

    printf("\n\nProcess\tAT\tBT\tPR\tCT\tWT\tTAT\n");
    for(int i = 0; i < n; i++)
        printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
               i+1, at[i], bt[i], pr[i], ct[i], wt[i], tat[i]);

    printf("\nAverage WT = %.2f", total_wt/n);
    printf("\nAverage TAT = %.2f\n", total_tat/n);

    return 0;
}
```

```c
#include <stdio.h>
#include <limits.h>

int main() {
    int n;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    int at[n], bt[n], pr[n], rt[n];
    int ct[n], wt[n], tat[n];

    printf("Enter Arrival Time, Burst Time and Priority:\n");
    for(int i = 0; i < n; i++) {
        printf("P%d AT: ", i+1);
        scanf("%d", &at[i]);
        printf("P%d BT: ", i+1);
        scanf("%d", &bt[i]);
        printf("P%d Priority: ", i+1);
        scanf("%d", &pr[i]);
        rt[i] = bt[i];
    }

    int time = 0, completed = 0;
    int prev = -1;
    float total_wt = 0, total_tat = 0;

    printf("\nGantt Chart:\n|");

    while(completed < n) {
        int idx = -1;
        int highest_pr = INT_MAX;

        for(int i = 0; i < n; i++) {
            if(at[i] <= time && rt[i] > 0) {
                if(pr[i] < highest_pr) {
                    highest_pr = pr[i];
                    idx = i;
```